I. Cycles through 1 for the $qn + 1$ map: no cycles of length 3, and constraints on longer cycles

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Abstract

For the $qn + 1$ map with q an odd integer at least 3, we prove that 1 never lies on a cycle with exactly three distinct odd numbers. This settles the case of length 3 in a conjecture of Santos (2020) that 1 lies on a cycle only for $q = 5$ and for Mersenne numbers $q = 2^e - 1$. We found that for every such q at least 2^{22001}, a lower bound and an upper bound cannot hold at the same time. Bauer and Bennett (2018) give an irrationality measure whose exponent, about 1.61487, lies just below the golden ratio, the threshold at which these bounds conflict. Every smaller q is ruled out by a direct computation.

1 Introduction

This paper studies cycles of the $qn+1$ map through 1. It is followed by the broader paper [9], which studies cycles of the same map for any starting value.

Fix an odd integer $q \geq 3$. For an odd positive integer n, let v be the largest integer such that 2^v divides $qn + 1$. The map C sends n to

$$C(n) = \frac{qn + 1}{2^v}.$$

The orbit of n is the sequence n, $C(n)$, $C(C(n))$, …. When $q = 3$, C is the odd-to-odd or accelerated form of the Collatz map, and the Collatz conjecture, posed by Lothar Collatz in 1937 and still open, asserts that every orbit eventually reaches 1. For $q = 3$ the number 1 is fixed, since $3 \cdot 1 + 1 = 4$. For larger q, heuristic arguments suggest that most orbits grow without bound and that cycles are rare, although no orbit has been proved to diverge.

We ask for which q the number 1 lies on a cycle of C. We classify the answers by the length ℓ of the cycle through 1, that is, the number of distinct odd numbers on it.

For $\ell = 1$ the cycle is $1 \to 1$, which happens exactly when $q + 1$ is a power of 2, that is $q = 2^e - 1$ (Proposition 15 of Santos (2020)). For $\ell = 2$, the only q is 5, with $1 \to 3 \to 1$ (Proposition 16 of Santos (2020); a short proof also follows from the theorem of Beukers (1980/81) on $x^2 + D = 2^n$, stated as Theorem 1.8 of Bauer and Bennett (2002)). Besides the Mersenne family $1 \to 1$, the only known cycle through 1 is $1 \to 3 \to 1$ for $q = 5$.

Santos's conjecture (2020). 1 lies on a cycle of C only for $q = 5$ and $q = 2^e - 1$.

Theorem 1. No odd $q \geq 3$ has 1 on a cycle of C of length 3.

Outline of Theorem 2 (golden-ratio criterion). Fix $\ell \geq 2$, and suppose the roots of 2 that this length produces have an approximation exponent below the golden ratio. Then any q with 1 on a cycle of C of length ℓ lies below a bound we can compute. Checking every candidate up to that bound finds every such q, or shows that none exist. The precise statement is in Section 6.

Theorem 3 (finiteness through 1). For each $\ell \geq 2$, only finitely many odd q have 1 on a cycle of C of length ℓ.

Theorem 3 is ineffective: it rests on the S-unit form of Ridout's theorem in Section 7, and gives no bound on q. (A finiteness result is called effective if its proof gives an explicit bound on any exception, so that a finite computation can find them all, and ineffective if it gives no such bound.) Theorem 1 is effective and is completed by computation. The proof of Theorem 3, in Section 7, uses only the S-unit form of Ridout's theorem and Theorem 2. Cycles that need not visit 1 are treated in the companion paper [9]. For a cycle of length 2 or 3 through a fixed odd element, that paper gives a bound on q depending only on the element (Theorem 3), and for the longer classes of Corollaries 12 to 14 the same bound is Corollary 4 of that paper.

2 Definitions and notation

Throughout, $q \geq 3$ is an odd integer and C is the map defined in Section 1.

The 2-adic valuation. For a nonzero integer n, $v_2(n)$ is the largest integer v such that 2^v divides n.

Cycles. A cycle of C of length ℓ consists of ℓ distinct odd positive integers. The smallest element is x_0. Applying C repeatedly, starting at x_0, produces

$$x_0 \to x_1 \to \cdots \to x_{\ell-1} \to x_0.$$

Cycles through 1. A cycle of length ℓ through 1 is a cycle of C of length ℓ with $x_0 = 1$. It is written

$$1 = x_0 \to x_1 \to \cdots \to x_{\ell-1} \to 1.$$

Halvings. The halving counts $k_1, \ldots, k_\ell$ of the cycle are defined by

$$qx_{i-1} + 1 = 2^{k_i} x_i \qquad (1 \leq i \leq \ell,\ x_\ell = x_0),$$

so $k_i = v_2(q \cdot x_{i-1} + 1)$. The total number of halvings is $h = k_1 + \cdots + k_\ell$, and $S_i = k_1 + \cdots + k_i$.

The window. On a cycle of length ℓ through 1 with h halvings in all, write $2^{h/\ell} = q + \delta$, so δ is the fractional part of $2^{h/\ell}$. The window is a lower bound and an upper bound on δ.

Near misses. For an element x_i other than 1 on a cycle of length ℓ through 1, set $n_i = ih - \ell S_i$. The near miss of x_i is $2^{n_i/\ell}$. The relative error ρ_i is defined by

$$x_i = 2^{n_i/\ell}(1 + \rho_i).$$

Approximation exponents. Let θ be irrational and let $\mathcal{D}$ be a set of positive integers. Then θ has effective approximation exponent μ with respect to $\mathcal{D}$ when there are explicit constants $c > 0$ and d_0 such that

$$\left|\theta - \frac{p}{d}\right| > c \cdot d^{-\mu}$$

for every integer p and every $d \in \mathcal{D}$ with $d \geq d_0$. Write $\mathcal{D}_\ell$ for the set of denominators 2^j and $\ell \cdot 2^j$, where j is a nonnegative integer.

The golden ratio. $\varphi = (1 + \sqrt{5})/2$.

The order of 2. For an odd positive integer m, the order of 2 modulo m is the smallest positive integer d such that $2^d \equiv 1 \pmod{m}$.

Other conventions. An implied constant in O, $\ll$, $\gg$, or $\asymp$ depends only on ℓ. In a heuristic remark, $\lesssim$ and $\gtrsim$ mean at most or at least roughly, ignoring constants.

The three steps of length 3. On a cycle of length 3 through 1, the element after 1 is $x = x_1$. The element after x is $y = x_2$, so the cycle is $1 \to x \to y \to 1$. The three steps are $q + 1 = 2^{k_1}x$, $qx + 1 = 2^{k_2}y$, and $qy + 1 = 2^{k_3}$.

3 The structure of a cycle of length 3

Lemma 1. $k_3 \geq k_1 + 3$, and $y \equiv 1 \pmod{2^{k_1}}$.

Proof. The odd elements other than 1 are at least 3, so $x \geq 3$ and $y \geq 3$. The first step is $q + 1 = 2^{k_1}x$, and therefore $q + 1 \geq 3 \cdot 2^{k_1}$, that is $q \geq 3 \cdot 2^{k_1} - 1$. The third step is $2^{k_3} = qy + 1$, so

$$2^{k_3} \geq (3 \cdot 2^{k_1} - 1) \cdot 3 + 1 = 9 \cdot 2^{k_1} - 2.$$

Since $2^{k_1} \geq 2$, we have $9 \cdot 2^{k_1} - 2 \geq 8 \cdot 2^{k_1} = 2^{k_1+3}$, and hence $k_3 \geq k_1 + 3$. From $q + 1 = 2^{k_1}x$ we also have $q \equiv -1 \pmod{2^{k_1}}$. Thus $qy + 1 \equiv 1 - y \pmod{2^{k_1}}$. But $qy + 1 = 2^{k_3}$ is divisible by 2^{k_1}, because $k_3 \geq k_1$, so $1 - y \equiv 0 \pmod{2^{k_1}}$, that is $y \equiv 1 \pmod{2^{k_1}}$. □

Lemma 2 (cycle equation).

$$2^h = q^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2}.$$

Proof. Since $h = k_1 + k_2 + k_3$,

$$2^h = 2^{k_1+k_2} \cdot 2^{k_3}.$$

The third step is $qy + 1 = 2^{k_3}$, so

$$2^h = 2^{k_1+k_2}(qy + 1) = 2^{k_1}(q \cdot 2^{k_2}y + 2^{k_2}).$$

The second step is $qx + 1 = 2^{k_2}y$, and therefore

$$2^h = 2^{k_1}(q(qx + 1) + 2^{k_2}) = 2^{k_1}(q^2x + q + 2^{k_2}) = q^2(2^{k_1}x) + 2^{k_1}q + 2^{k_1+k_2}.$$

The first step is $q+1 = 2^{k_1}x$, so

$$2^h = q^2(q+1) + 2^{k_1}q + 2^{k_1+k_2} = q^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2}.$$

□

Lemma 3 (cube window). $q^3 < 2^h < (q+1)^3$. Hence 3 does not divide h and $q = \lfloor 2^{h/3} \rfloor$.

Proof. The lower bound is immediate from Lemma 2. Since $2^{k_1+k_2} = (q(q+1) + 2^{k_1})/y \leq (q+1)(3q+1)/9$ by $y \geq 3$, and $2^{k_1} \leq (q+1)/3$,

$$2^h - q^3 \leq q^2 + \frac{q(q+1)}{3} + \frac{(q+1)(3q+1)}{9} = \frac{15q^2+7q+1}{9} < (q+1)^3 - q^3.$$

A power of 2 strictly between consecutive cubes is not a cube, so $3 \nmid h$. □

Lemma 4 (2-adic valuation of $x-1$). Let $t = v_2(x-1)$ and $s = (x-1)/2^t$. Then either $t = 2k_1$, or $t < k_1$ and $k_1 = t + v_2(s+1)$, or $t = k_1$.

Proof. The first step is $q+1 = 2^{k_1}x$, so $q = 2^{k_1}x - 1$.

Substitute $q = 2^{k_1}x - 1$ into the second step $2^{k_2}y = qx + 1$:

$$2^{k_2}y = (2^{k_1}x - 1)x + 1 = 2^{k_1}x^2 - x + 1 = 2^{k_1}x^2 - (x-1).$$

Suppose $t > k_1$. Since x is odd, $v_2(2^{k_1}x^2) = k_1$, while $v_2(x-1) = t > k_1$, so

$$2^{k_1}x^2 - (x-1) = 2^{k_1}(x^2 - 2^{t-k_1}s),$$

and $x^2 - 2^{t-k_1}s$ is odd, since x^2 is odd and $2^{t-k_1}s$ is even. Thus the 2-adic valuation of $2^{k_2}y$ is k_1. Since y is odd, $k_2 = k_1$, so $2^{k_1}y = 2^{k_1}x^2 - (x-1)$.

Let $w = (x-1)/2^{k_1}$. Then w is even, since $t > k_1$, and $y = x^2 - w$. Substituting $x = 1 + 2^{k_1}w$,

$$qy + 1 = 2^{k_1} + w(1 - 3 \cdot 2^{k_1}) + 2^{2k_1}w(3 - 2w) + 3 \cdot 2^{3k_1}w^2 + 2^{4k_1}w^3.$$

The terms $2^{2k_1}w(3-2w)$, $3 \cdot 2^{3k_1}w^2$, and $2^{4k_1}w^3$ are divisible by $2^{2k_1+v_2(w)}$, and $1 - 3 \cdot 2^{k_1}$ is odd, so $v_2(w(1 - 3 \cdot 2^{k_1})) = v_2(w)$. If $v_2(w) \neq k_1$, then $v_2(qy+1) = \min(k_1, v_2(w)) \leq k_1$, contradicting Lemma 1. Hence $v_2(w) = k_1$, that is $t = 2k_1$.

Suppose $t < k_1$. Then $2^{k_2}y = 2^{k_1}x^2 - 2^t s = 2^t(2^{k_1-t}x^2 - s)$. The factor $2^{k_1-t}x^2 - s$ is odd, since $2^{k_1-t}x^2$ is even and s is odd, so $k_2 = t$, $y = 2^{k_1-t}x^2 - s$, and

$$qy + 1 = (2^{k_1} - 1)\,2^{k_1-t}x^2 + (1+s) - 2^{k_1}s + 2^{k_1+t}sy.$$

Now $2^{k_1}s$ and $2^{k_1+t}sy$ are divisible by 2^{k_1}, and $v_2((2^{k_1}-1)\,2^{k_1-t}x^2) = k_1 - t$, since $2^{k_1} - 1$ and x are odd. If $v_2(s+1) \neq k_1 - t$, then $v_2(qy+1) < k_1$, contradicting Lemma 1. □

Lemma 5. $2^{k_1} \le 2(x-1)$ and $x > \sqrt{q/2}$.

Proof. In the three cases of Lemma 4: $2^{2k_1} = 2^t \le x-1$; or $2^{k_1} = 2^t \cdot 2^{v_2(s+1)} \le 2^t(s+1) = (x-1)+2^t \le 2(x-1)$; or $2^{k_1} = 2^t \le x-1$. The first step is $q+1 = 2^{k_1}x$, so

$$q+1 \le 2x(x-1) = 2x^2 - 2x.$$

Hence $q < 2x^2$, that is $x > \sqrt{q/2}$. □

Lemma 6 (consecutive triples). Let $a \to b \to c$ be consecutive elements of any orbit of C, with $qa+1 = 2^J b$ and $qb+1 = 2^K c$. Then

$$2^K ac - 2^J b^2 = a - b,$$

so if $a \ne b$ then $|a-b| \ge 2^{\min(J,K)}$. Consequently, if $J \le K$ then $b > q$ when $b < a$, and $b^2 > qa$ when $b > a$.

Proof. Substitute $q = (2^K c - 1)/b$ into $qa+1 = 2^J b$ and multiply by b. If $J \le K$, then $|a-b| \ge 2^J = (qa+1)/b$; if $b > a$ this gives $b^2 > b(b-a) \ge qa+1$, and if $b < a$ it gives $b(a-b) \ge qa+1$, so $b > q$. □

Lemma 7. $y > x$ and $y^2 > qx$. In particular $2^{k_2} < q+1$, so $2^{k_1+k_2} \le 2^{k_1}q$.

Proof. Apply Lemma 6 to $x \to y \to 1$, so $J = k_2$ and $K = k_3$. If $k_3 < k_2$, then $|x-y| \ge 2^{k_3} = qy+1 > y$, which forces $x > qy > q$, impossible as $x \le (q+1)/2$. So $k_2 \le k_3$. If $y < x$, Lemma 6 gives $y > q > x$, a contradiction. Hence $y > x$ and $y^2 > y(y-x) \ge qx+1$. Finally $2^{k_2} = (qx+1)/y < (qx+1)/x < q+1$, and 2^{k_2} is even while q is odd. □

4 The window and the near miss for x

Lemma 8 (the window). Let $N = 27 \cdot 2^h - (3q+1)^3$. Then

$$N = 9q(3 \cdot 2^{k_1} - 1) + 27 \cdot 2^{k_1+k_2} - 1, \qquad 0 < N < 54 \cdot 2^{k_1} q, \qquad 0 < \delta - \frac{1}{3} \le \frac{N}{81q^2} < \frac{2^{k_1+1}}{3q}.$$

Proof. The formula for N follows from Lemma 2 by expanding $(3q+1)^3$. Since $2^{k_1} \ge 2$, $N \ge 45q - 1 > 0$, and $2^{k_1+k_2} \le 2^{k_1}q$ (Lemma 7) gives $N < 54 \cdot 2^{k_1}q$. Finally $N = 27[(q+\delta)^3 - (q+1/3)^3]$; as $N > 0$ we have $\delta > 1/3$, and then $N \ge 81q^2(\delta - 1/3)$. □

Lemma 9 (near miss for x). Let $f = h - 3k_1$. Then $3 \nmid f$, and

$$\left|x^3 - 2^f\right| \le \frac{3.2\,x^2}{2^{k_1}}, \qquad \left|x - 2^{f/3}\right| \le \frac{1.6}{2^{k_1}}.$$

Proof. The first step is $q+1 = 2^{k_1}x$, so $2^{3k_1}x^3 = (q+1)^3$.

Let $E = 2q^2 + (3-2^{k_1})q + 1 - 2^{k_1+k_2}$. Lemma 2 gives $2^h = q^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2}$, so

$$2^{3k_1}x^3 - 2^h = (q+1)^3 - 2^h = 2q^2 + (3-2^{k_1})q + 1 - 2^{k_1+k_2} = E.$$

Since $2^{k_1} \geq 2$, either $2^{k_1} = 2$ and $|3-2| = 1$, or $2^{k_1} \geq 4$ and $|3-2^{k_1}| = 2^{k_1} - 3$. In both cases $|3-2^{k_1}| \leq 2^{k_1}$. Since $q < 2^{k_1}x$ and, by Lemma 7, $2^{k_1+k_2} \leq 2^{k_1}q < 2^{2k_1}x$,

$$|E| < 2 \cdot 2^{2k_1}x^2 + 2 \cdot 2^{2k_1}x + 1 = 2^{2k_1}x^2 \left(2 + \frac{2}{x} + \frac{1}{2^{2k_1}x^2}\right).$$

Since $x \geq 3$ and $2^{k_1} \geq 2$,

$$2 + \frac{2}{x} + \frac{1}{2^{2k_1}x^2} \leq 2 + \frac{2}{3} + \frac{1}{36} = \frac{97}{36} < 3.2,$$

so $|E| < 3.2 \cdot 2^{2k_1}x^2$. Dividing by 2^{3k_1} gives $\left|x^3 - 2^f\right| < 3.2\,x^2/2^{k_1}$, and in particular

$$\left|x^3 - 2^f\right| \leq \frac{3.2\,x^2}{2^{k_1}}.$$

Since $f = h - 3k_1$, one has $f \equiv h \pmod 3$. Lemma 3 says that 3 does not divide h, so 3 does not divide f.

$$\left|x^3 - 2^f\right| = \left|x - 2^{f/3}\right|(x^2 + x \cdot 2^{f/3} + 2^{2f/3}).$$

Suppose $\left|x - 2^{f/3}\right| \geq 1$. For every real number z, $\left|x^3 - z^3\right|$ increases as z moves away from x, so

$$\left|x^3 - 2^f\right| \geq \min(x^3 - (x-1)^3,\ (x+1)^3 - x^3) = 3x^2 - 3x + 1,$$

because $(x+1)^3 - x^3 = 3x^2 + 3x + 1 > 3x^2 - 3x + 1$. Since $x \geq 3$, one has $3x^2 - 3x + 1 \geq 2x^2$. Since $2^{k_1} \geq 2$, one has $3.2\,x^2/2^{k_1} \leq 1.6x^2 < 2x^2$. So $\left|x^3 - 2^f\right| \geq 2x^2$ and $\left|x^3 - 2^f\right| < 2x^2$, which is impossible.

Hence $\left|x - 2^{f/3}\right| < 1$, so $2^{f/3} > x - 1$. Since $x \geq 3$, one has $x - 1 \geq 2x/3$, and therefore $2^{f/3} > 2x/3$. Then

$$x^2 + x \cdot 2^{f/3} + 2^{2f/3} > x^2 + \frac{2x^2}{3} + \frac{4x^2}{9} = \frac{19}{9}x^2 > 2x^2,$$

so

$$\left|x - 2^{f/3}\right| = \frac{\left|x^3 - 2^f\right|}{x^2 + x \cdot 2^{f/3} + 2^{2f/3}} < \frac{3.2\,x^2}{2^{k_1} \cdot 2x^2} = \frac{1.6}{2^{k_1}}.$$

In particular $\left|x - 2^{f/3}\right| \leq 1.6/2^{k_1}$. □

5 Proof of Theorem 1

We use the following effective irrationality measure, which is the inequality obtained in the proof of Theorem 2.1 of Bauer and Bennett (2018) from their inequality (5.3), Lemma 4.1 and Proposition 5.1, before they round it to $2^{-1.62k}$ in (2.1).

Theorem (Bauer–Bennett (2018)). Let $j \geq 11000$, $r \in \{1,2\}$, $s \in \{1,3\}$ and $n = 3j + r$. For every integer p,

$$\left|2^{r/3} - \frac{p}{s\,2^j}\right| \geq \left|1 - \frac{p}{s\,2^{n/3}}\right| > 2^{-5} \cdot 16166467234^{-(n+77)/63}.$$

The factor 2^{-5} is a weakening of the contour bound behind their inequality (5.3). The denominator of (5.3) contributes $1/(1.26 \cdot 2^{14/3}) = 2^{-5.00009\ldots}$, and the contour estimate for $|P_{n_1,\delta}|$ has constant $1.2528\ldots$ in place of 1.26, so that $1.2528\ldots \cdot 2^{14/3}$ has base-2 logarithm $4.9918 < 5$. Their displayed intermediate step omits the factor, though it is absorbed in their final statement. Per unit of j the exponent is $\mu = 3\log_2(16166467234)/63 \approx 1.61487$.

Proof of Theorem 1. Suppose $q \geq 2^{22001}$. Write $a = k_1$, $h = 3m + r$ and $f = h - 3a = 3j' + r'$, where $r, r' \in \{1,2\}$ by Lemmas 3 and 9. Lemma 5 gives $x > \sqrt{q/2} \geq 2^{11000}$. Since q is odd, $a \geq 1$, and Lemma 9 gives $|x - 2^{f/3}| \leq 0.8$. Also $2^{f/3} \leq 2^{j'+2/3}$. If $j' \leq 10999$, then $x \leq 2^{10999+2/3} + 0.8$. Now $(8/5)^3 = 512/125 > 4$, so $2^{2/3} < 8/5$, and $2^{11000} - (8/5)\,2^{10999} = (2/5)\,2^{10999} > 0.8$. Thus $x < 2^{11000}$, a contradiction, and $j' \geq 11000$. Likewise $(5/4)^3 = 125/64 < 2$, so $2^{1/3} > 5/4$ and $2^{j'}/4 > 0.8$, which gives $x \geq 2^{j'+1/3} - 0.8 > 2^{j'} + 2^{j'}/4 - 0.8 > 2^{j'}$. Hence $x \geq 2^{j'} + 1$ and $q = 2^a x - 1 \geq 2^{a+j'} + 2^a - 1 \geq 2^{a+j'} + 1$. From Lemma 3, $2^h > q^3$, so $2^{h/3} > q > 2^{a+j'}$ and $m + r/3 > a + j'$. As $r \leq 2$, this forces $m \geq a + j' \geq 11000$.

1. **The window.** Apply Theorem (Bauer–Bennett (2018)) with $s = 3$, $p = 3q + 1$, $j = m$. Here $1 - p/(3 \cdot 2^{h/3}) = (\delta - 1/3)/2^{h/3}$, so $\delta - 1/3 > 2^m \cdot 2^{-5} \cdot 16166467234^{-(h+77)/63}$. With Lemma 8 and $q \geq 2^m$,
$$2^a > 1.5 \cdot 2^{2m-5} \cdot 16166467234^{-(h+77)/63}.$$

2. **The near miss for x.** Apply Theorem (Bauer–Bennett (2018)) with $s = 1$, $p = x$, $j = j'$. By Lemma 9, $|1 - x/2^{f/3}| \leq 1.6/(2^a \cdot 2^{f/3})$, so
$$2^a < 1.6 \cdot 2^{5-f/3} \cdot 16166467234^{(f+77)/63}.$$

3. **Combining.** With $h = f + 3a$, $m \geq a + j'$ and $j' \geq (f-2)/3$, step 1 gives $a > 0.20878\,f - 76.8$ and step 2 gives $a < 0.20496\,f + 47.2$. These are incompatible once $f \geq 32{,}461$, that is $j' \geq 10{,}820$.

So any cycle of length 3 through 1 has $q < 2^{22001}$. Lemma 3 gives $2^h < (q+1)^3$. Since q is an integer, $q + 1 \leq 2^{22001}$, so $2^h < 2^{66{,}003}$ and $h < 66{,}003$. Each such h gives a single candidate $q = \lfloor 2^{h/3} \rfloor$. Appendix E tests every $h \leq 200{,}000$ and finds no cycle, and the digit scan there rules out every such cycle with $q < 2^{1{,}071{,}999{,}953}$. □

Why the golden ratio appears. The proof squeezes x between two bounds, both governed by the exponent μ in Theorem (Bauer–Bennett (2018)).

- **The window keeps x small.** By Lemma 8, $2^{h/3}$ exceeds $q + 1/3$ by about $1/(3x)$, so $(3q+1)/(3 \cdot 2^m)$ approximates $2^{r/3}$ with error about $1/(xq)$. Theorem (Bauer–Bennett (2018)) forbids errors below about $q^{-\mu}$, so $x \lesssim q^{\mu-1}$.

- **The near miss for x keeps x large.** By Lemma 9, $x/2^{j'}$ approximates $2^{r'/3}$ with error about $1/q$, where $2^{j'} \asymp x$. Theorem (Bauer–Bennett (2018)) forbids errors below about $x^{-\mu}$, so $x \gtrsim q^{1/\mu}$.

Both bounds can hold only if $1/\mu \le \mu - 1$, that is $\mu^2 - \mu - 1 \ge 0$, whose positive root is $\varphi = (1+\sqrt{5})/2$. With $\mu = 1.61487$ the bounds read $q^{0.6192} \lesssim x \lesssim q^{0.6149}$, which are incompatible once q is large; the explicit constants in steps 1 to 3 make this precise for $q \ge 2^{22001}$. The two bounds are tied together by $q + 1 = 2^{k_1} x$: a large x makes the window's approximation too good, while a small x forces 2^{k_1} to be large and makes x's own approximation too good.

The threshold φ comes from this argument; the exponent comes from Bauer and Bennett, who needed only an exponent below 2 for their own purposes. Their stated exponent 1.62 lies just above φ and would leave room for x; the unrounded 1.61487 lies just below it.

6 Longer cycles

Lemma 10 (the ℓ-th power window). For a cycle of length ℓ through 1, let $\sigma = \sum_{i=1}^{\ell-1} 1/x_i$. Then

$$2^h = (q+1) \prod_{i=1}^{\ell-1} \left(q + \frac{1}{x_i} \right).$$

If $\ell \ge 2$, then $q^\ell < 2^h < (q+1)^\ell$, ℓ does not divide h, and $q = \lfloor 2^{h/\ell} \rfloor$. Also $\delta - 1/\ell = \sigma/\ell + O(1/q)$.

Proof. For $1 \le i \le \ell$, the step from x_{i-1} is $2^{k_i} x_i = q x_{i-1} + 1$. Since $h = k_1 + \cdots + k_\ell$, $x_0 = 1$ and $x_\ell = 1$,

$$2^h = \prod_{i=1}^{\ell} \frac{q x_{i-1} + 1}{x_i} = (q+1) \prod_{i=1}^{\ell-1} \left(q + \frac{1}{x_i} \right).$$

When $\ell = 1$, the product $\prod_{i=1}^{0}$ is empty, and $2^h = q + 1$.

Suppose $\ell \ge 2$. Each x_i with $1 \le i \le \ell - 1$ is an odd integer at least 3, so $q < q + 1/x_i < q + 1$. Therefore

$$q^\ell < (q+1)\, q^{\ell-1} < (q+1) \prod_{i=1}^{\ell-1} \left(q + \frac{1}{x_i} \right) < (q+1)^\ell.$$

The product $(q+1) \prod_{i=1}^{\ell-1} (q + 1/x_i)$ equals 2^h, so $q^\ell < 2^h < (q+1)^\ell$. A power of 2 strictly between consecutive ℓ-th powers is not an ℓ-th power, so ℓ does not divide h, and $q = \lfloor 2^{h/\ell} \rfloor$.

Since $2^{h/\ell} = q + \delta$, one has $(q+\delta)^\ell = 2^h$. Dividing by q^ℓ gives

$$\left(1 + \frac{\delta}{q} \right)^\ell = \left(1 + \frac{1}{q} \right) \prod_{i=1}^{\ell-1} \left(1 + \frac{1}{q x_i} \right).$$

Expanding the product, $\prod_{i=1}^{\ell-1} (1 + 1/(q x_i)) = 1 + \sigma/q + O(1/q^2)$: the sum of the factors $1/(q x_i)$ is σ/q, and every product of two or more such factors is $O(1/q^2)$, with the constant depending only on ℓ. Thus

$$\left(1 + \frac{1}{q} \right) \left(1 + \frac{\sigma}{q} + O(1/q^2) \right) = 1 + \frac{1+\sigma}{q} + O(1/q^2).$$

Since $0 < \delta < 1$, the binomial expansion gives $(1+\delta/q)^\ell = 1 + \ell\delta/q + O(1/q^2)$. From $1 + \ell\delta/q + O(1/q^2) = 1+(1+\sigma)/q+O(1/q^2)$ one has $\ell\delta/q = (1+\sigma)/q+O(1/q^2)$, so $\delta - 1/\ell = \sigma/\ell + O(1/q)$. $\square$

Lemma 11 (every element is a near miss). For $1 \le i \le \ell - 1$, let

$$\varepsilon_i = \frac{1}{q} + \sum_{u=1}^{i-1} \frac{1+\varepsilon_u}{q x_u}.$$

Then $x_i\, 2^{S_i} = q^i(1+\varepsilon_i)$. Since $n_i = ih - \ell S_i$, one has $x_i = 2^{n_i/\ell}(1+\rho_i)$. For $q \ge \ell$,

$$-\frac{i}{q+1} < \rho_i \le \frac{2i+1}{3q}.$$

In particular $|\rho_i| \le (\ell-1)/q$. If $q \ge \ell$ and ℓ divides n_i, then $x_i \ge q/(\ell-1) - 1$.

Proof. For $i = 1$, one has $\varepsilon_1 = 1/q$ and $x_1 2^{k_1} = q + 1 = q(1 + 1/q)$.

Suppose $i \ge 2$ and $x_{i-1} 2^{S_{i-1}} = q^{i-1}(1+\varepsilon_{i-1})$. The step from x_{i-1} is $2^{k_i} x_i = q x_{i-1} + 1$, and $S_i = S_{i-1} + k_i$, so

$$x_i 2^{S_i} = 2^{S_{i-1}}(q x_{i-1} + 1) = q^i(1+\varepsilon_{i-1}) + 2^{S_{i-1}}.$$

The inductive equation gives $2^{S_{i-1}}/q^i = (1+\varepsilon_{i-1})/(q x_{i-1})$. The definition of ε_i gives $\varepsilon_i = \varepsilon_{i-1} + (1+\varepsilon_{i-1})/(q x_{i-1})$. Therefore $x_i 2^{S_i} = q^i(1+\varepsilon_i)$.

Since $n_i = ih - \ell S_i$ and $x_i = 2^{n_i/\ell}(1+\rho_i)$,

$$1 + \rho_i = \frac{x_i 2^{S_i}}{2^{ih/\ell}} = \left(\frac{q}{q+\delta}\right)^i (1+\varepsilon_i),$$

where $2^{h/\ell} = q + \delta$. For $\ell \ge 2$, Lemma 10 gives $0 < \delta < 1$.

Now $\varepsilon_1 = 1/q = (2 \cdot 1 + 1)/(3q)$. If $\varepsilon_u \le 1$ for every $u < i$, then $(1+\varepsilon_u)/(q x_u) \le 2/(3q)$, because $x_u \ge 3$, and

$$\varepsilon_i \le \frac{1}{q} + \frac{2}{q} \cdot \frac{i-1}{3} = \frac{2i+1}{3q}.$$

For $q \ge \ell$ and $i \le \ell - 1$, one has $(2i+1)/(3q) \le (2\ell-1)/(3\ell) < 1$, so $\varepsilon_i \le 1$. Thus $\varepsilon_i \le (2i+1)/(3q)$ for every $i \le \ell - 1$. Since $0 < \delta < 1$, one has $\big(q/(q+1)\big)^i < \big(q/(q+\delta)\big)^i$, and $\big(q/(q+1)\big)^i = \big(1 - 1/(q+1)\big)^i \ge 1 - i/(q+1)$. Also $\big(q/(q+\delta)\big)^i \le 1$ and $1 + \varepsilon_i \le 1 + (2i+1)/(3q)$, so

$$1 - \frac{i}{q+1} \le \left(1 - \frac{1}{q+1}\right)^i < \left(\frac{q}{q+\delta}\right)^i (1+\varepsilon_i) \le 1 + \frac{2i+1}{3q}.$$

Here $\big(q/(q+\delta)\big)^i(1+\varepsilon_i) = 1 + \rho_i$, so $-i/(q+1) < \rho_i \le (2i+1)/(3q)$. Also $(2i+1)/(3q) \le (\ell-1)/q$ and $i/(q+1) < (\ell-1)/q$, so $|\rho_i| \le (\ell-1)/q$.

Suppose $q \ge \ell$ and ℓ divides n_i, so n_i/ℓ is an integer. If $n_i \ge 0$, then $2^{n_i/\ell}$ is a positive integer power of 2, hence 1 or even, while x_i is odd and at least 3, so $|x_i - 2^{n_i/\ell}| \ge 1$. If $n_i < 0$, then

$2^{n_i/\ell} \le 1/2$ and $x_i \ge 3$, so $|x_i - 2^{n_i/\ell}| \ge 5/2$. In either case $|\rho_i| = |x_i - 2^{n_i/\ell}| \cdot 2^{-n_i/\ell} \ge 2^{-n_i/\ell}$. Since $2^{n_i/\ell} = x_i/(1+\rho_i)$ and $|\rho_i| \le (\ell-1)/q < 1$,

$$2^{-n_i/\ell} = \frac{1+\rho_i}{x_i} \ge \frac{1-|\rho_i|}{x_i}.$$

Hence $|\rho_i| \ge (1-|\rho_i|)/x_i$, so $x_i \ge 1/|\rho_i| - 1 \ge q/(\ell-1) - 1$. □

The outline in the introduction is the case $R = \{1, \dots, \ell-1\}$.

Theorem 2. Let $\ell \ge 2$, let $R \subseteq \{1, \dots, \ell-1\}$, and let $\varphi = (1+\sqrt{5})/2$. Suppose that there are real numbers $0 < \mu < \varphi$, $c > 0$ and j_0 such that

$$\Big|2^{r/\ell} - \frac{p}{s \cdot 2^j}\Big| > c \cdot 2^{-\mu j}$$

for all $r \in R$, $s \in \{1, \ell\}$, $p \in \mathbb{Z}$ and integers $j \ge j_0$. Then there is a number Q, computable from ℓ, μ, c and j_0, with the following property. If $q > Q$ is odd and $1 = x_0 \to x_1 \to \cdots \to x_{\ell-1} \to 1$ is a cycle of C of length ℓ, with $q \cdot x_{i-1} + 1 = 2^{k_i} \cdot x_i$ for $1 \le i \le \ell$ (where $x_\ell = 1$), then some multiple of $h = k_1 + \cdots + k_\ell$ has a nonzero residue modulo ℓ outside R.

For $R = \{1, \dots, \ell-1\}$ the conclusion is that no $q > Q$ has 1 on a cycle of length ℓ. For $\ell = 4$ and $R = \{2\}$, the multiples of h avoid the residues 1 and 3 exactly when h is even, so the conclusion is that h is odd.

Proof of Theorem 2. Suppose that every multiple of h has its nonzero residues modulo ℓ in R. Write $h = \ell m + r$ with $r \in R$, and assume $q \ge 2(\ell-1)$, so Lemma 11 gives $|\rho_i| \le 1/2$. The window uses $2^{r/\ell}$, and x_i uses $2^{n_i/\ell}$ with $n_i \equiv ir \pmod{\ell}$, so every root below has its residue in R. Set

$$A = 2^{1/\ell - 1}\Big(\frac{c}{2(\ell-1)}\Big)^{1/\mu}, \qquad \gamma = 1 - \mu + \frac{1}{\mu} > 0,$$

the last because $\mu^2 - \mu - 1 < 0$.

If $q < 2^{j_0+1}$, stop: this is already a bound. If $q \ge 2^{j_0+1}$, then $2^m = (q+\delta)/2^{r/\ell} > q/2 \ge 2^{j_0}$, so $m \ge j_0$ and the hypothesis applies to the window.

$$\Big|\delta - \frac{1}{\ell}\Big| = 2^m \Big|2^{r/\ell} - \frac{\ell q + 1}{\ell \cdot 2^m}\Big| > c \cdot 2^{(1-\mu)m}.$$

If $\mu > 1$ then $2^m < q+1$ gives $|\delta - 1/\ell| > c(q+1)^{1-\mu} \ge c\, 2^{1-\mu} q^{1-\mu}$. If $\mu \le 1$ then $2^m > q/2$ gives $|\delta - 1/\ell| > c(q/2)^{1-\mu} \ge c/2$.

For each x_i with $\ell \nmid n_i$, write $n_i = \ell j + r'$ and $x_i/2^j = 2^{r'/\ell}(1+\rho_i)$, so $|2^{r'/\ell} - x_i/2^j| = 2^{r'/\ell}|\rho_i|$. Also $x_i/3 \le 2^j \le 2x_i$. If $j < j_0$, then $x_i < 3 \cdot 2^{j_0}$. The finitely many triples (r', j, x) with $r' \in R$, $0 \le j < j_0$ and x odd, $3 \le x < 3 \cdot 2^{j_0}$, each have a positive $|\rho| = |x \cdot 2^{-j-r'/\ell} - 1|$, and Lemma 11 forces $q \le (\ell-1)/|\rho|$. Let Q_0 be the maximum of those finitely many values. If $j \ge j_0$, the hypothesis and $|\rho_i| \le (\ell-1)/q$ give $c \cdot 2^{-\mu j} < 2(\ell-1)/q$, so

$$2^j > \Big(\frac{cq}{2(\ell-1)}\Big)^{1/\mu}, \qquad x_i \ge 2^{1/\ell - 1}\, 2^j \ge A\, q^{1/\mu}.$$

If $\ell \mid n_i$, then $x_i \geq q/(\ell-1) - 1 \geq q/(2(\ell-1))$. Thus every x_i is at least $Aq^{1/\mu}$ once q is large enough that $Aq^{1/\mu} \leq q/(2(\ell-1))$, and

$$\sigma \leq (\ell-1)A^{-1}q^{-1/\mu}.$$

Expanding the product in Lemma 10,

$$\Big(1+\frac{\delta}{q}\Big)^{\ell} = \Big(1+\frac{1}{q}\Big)\prod_{i=1}^{\ell-1}\Big(1+\frac{1}{qx_i}\Big).$$

The product is $1+\sigma/q+\rho$ with $0<\rho\leq(\sigma/q)^2e^{\sigma/q}$. Comparing with the binomial expansion of the left side, and using $\sigma\leq(\ell-1)/3$ and $q\geq\ell$,

$$\Big|\delta - \frac{1}{\ell} - \frac{\sigma}{\ell}\Big| \leq \frac{\sigma}{\ell q} + \frac{\rho(q+1)}{\ell} + \frac{\ell\, e^{\ell/q}}{q} \leq \frac{4\ell}{q}.$$

Hence $|\delta - 1/\ell| \leq A^{-1}q^{-1/\mu} + 4\ell/q$. For $\mu \geq 1$ the second term is at most $4\ell\, q^{-1/\mu}$, so the right-hand side is at most $(A^{-1}+4\ell)\, q^{-1/\mu}$.

If $\mu > 1$, the two bounds on $|\delta - 1/\ell|$ conflict once $c\, 2^{1-\mu}q^{1-\mu} > (A^{-1}+4\ell)\, q^{-1/\mu}$, that is

$$q > Q_* := \left(\frac{A^{-1}+4\ell}{c\, 2^{1-\mu}}\right)^{1/\gamma}.$$

If $\mu \leq 1$, they conflict once $c/2 > (A^{-1}+4\ell)\, q^{-1/\mu}$, that is $q > \big(2(A^{-1}+4\ell)/c\big)^{\mu}$; use this as Q_* instead.

Take $Q = \max(2(\ell-1),\ 2^{j_0+1},\ Q_0,\ Q_*)$. □

For $\ell = 3$, Theorem (Bauer–Bennett (2018)) supplies the hypothesis of Theorem 2 with $R = \{1,2\}$, $\mu \approx 1.61487$ and $j_0 = 11000$. So Theorem 1 is the case $\ell = 3$ of Theorem 2; Section 5 carries out this case with explicit constants.

Corollary 12. No odd q has 1 on a cycle of C of length 4 whose total number of halvings is $\equiv 2 \pmod 4$.

Proof. This is Theorem 2 with $\ell = 4$ and $R = \{2\}$, with explicit constants. Write $h = 4m+2$. Then $n_1 \equiv n_3 \equiv 2$ and $n_2 \equiv 0 \pmod 4$, so only $\sqrt{2}$ and powers of 2 occur, and the window approximates $\sqrt{2}$ by $(4q+1)/2^{m+2}$. Bauer and Bennett's Corollary 1.6 (2002) gives $|\sqrt{2} - p/2^j| > 2^{-1.48j}$ for every $j > 2$ with $j \notin \{3,7,8\}$.

- By Lemma 11, $x_2 \geq q/3.1$, and x_1, x_3 are near $\sqrt{2}\cdot 2^j$, so $4.3/q > (x/1.41)^{-1.48}$, giving $x_1, x_3 \geq (0.387q)^{0.6757}$.
- The measure gives $|\delta - 1/4| > 0.152(q+1)^{-0.48}$, while expanding the product in Lemma 10 explicitly gives $|\delta - 1/4| \leq \sigma/4 + 1.75/q \leq 0.95\, q^{-0.6757} + 2.53/q$.

For $q \geq 10^5$ every exponent involved exceeds 8, so Corollary 1.6 applies, and the two bounds are then incompatible (the inequalities already conflict from $q = 2 \times 10^4$). The search over h covers $\ell = 4$ for $q < 2^{25{,}000}$. □

Corollary 13. Let $\ell = 2^a$ with $a \geq 2$. Only finitely many odd q have 1 on a cycle of C of length ℓ whose total number of halvings is $\equiv \ell/2 \pmod{\ell}$, and a bound is computable from Bauer and Bennett's Corollary 1.6 (2002). For $\ell = 4$ every such q is ruled out by Corollary 12.

Proof. Write $h = \ell m + \ell/2$. Every multiple of h is congruent to 0 or $\ell/2$ modulo ℓ, so the only root that occurs is $2^{1/2}$. A denominator $s \cdot 2^j$ with $s \in \{1, \ell\}$ is a power of 2, because ℓ is. Bauer and Bennett's Corollary 1.6 (2002) therefore gives the hypothesis of Theorem 2 with $R = \{\ell/2\}$, $\mu = 1.48 < \varphi$, $c = 1$ and $j_0 = 9$: the finitely many excluded exponents $j \in \{1, 2, 3, 7, 8\}$ contribute only a constant factor depending on ℓ. The bound is then the Q in the proof of Theorem 2 for these values. The case $\ell = 4$ is Corollary 12. □

Corollary 14. Let $\ell = 3 \cdot 2^a$ with $a \geq 1$. Only finitely many odd q have 1 on a cycle of C of length ℓ whose total number of halvings is $\equiv \ell/3$ or $2\ell/3 \pmod{\ell}$, and a bound is computable from Theorem (Bauer–Bennett (2018)). The length 3 is Theorem 1, which rules out every such q.

Proof. Write $h = \ell m + r_0$ with $r_0 \in \{\ell/3,\, 2\ell/3\}$. Every multiple of h is congruent to 0, $\ell/3$, or $2\ell/3$ modulo ℓ, so the only irrational roots that occur are $2^{1/3}$ and $2^{2/3}$. In the proof of Theorem 2 the window fraction is $(\ell q + 1)/(\ell \cdot 2^m)$. Its numerator is coprime to ℓ, and $\ell \cdot 2^m = 3 \cdot 2^{m+a}$, so this is a fraction of the shape in Theorem (Bauer–Bennett (2018)) with $s = 3$ and exponent $m + a$. Each near miss is $x_i/2^j$ with x_i odd, so Theorem (Bauer–Bennett (2018)) applies with $s = 1$. Both give the exponent $\mu = 3\log_2(16166467234)/63 \approx 1.61487 < \varphi$ once the exponent in Theorem (Bauer–Bennett (2018)) is at least 11000. The shift by a, and the fixed factors in Theorem (Bauer–Bennett (2018)), contribute only a constant depending on ℓ. The fractions used in the proof of Theorem 2 therefore satisfy its hypothesis with $R = \{\ell/3,\, 2\ell/3\}$, this μ, a $c > 0$ read from Theorem (Bauer–Bennett (2018)), and $j_0 = 11000$. The bound is then the Q in the proof of Theorem 2 for these values. The length 3 is Theorem 1. □

7 Ineffective finiteness

Ridout's theorem (1958), S-unit form. Let α be an algebraic irrational, let S be a finite set of primes, and let $\varepsilon > 0$. Only finitely many reduced fractions p/Q with $Q > 0$ have every prime factor of Q in S and satisfy

$$\left|\alpha - \frac{p}{Q}\right| < Q^{-1-\varepsilon}.$$

This form gives no bound on the exceptional fractions. Since α is irrational, the finitely many fractions that meet it for a fixed α lie at a positive distance from α, and these distances have a positive minimum. This is why Theorem 3, which uses this form in place of an effective measure, gives no bound on q.

Theorem 3 (finiteness through 1). For each $\ell \geq 2$, only finitely many odd q have 1 on a cycle of C of length ℓ.

The argument below uses only the S-unit form of Ridout's theorem and Theorem 2.

Proof. Let $r \in \{1, \ldots, \ell - 1\}$ and let S consist of 2 and the prime factors of ℓ. Then $2^{r/\ell}$ is an algebraic irrational. A fraction $p/(s \cdot 2^j)$ with $s \in \{1, \ell\}$, written in lowest terms, has every prime factor of its denominator in S. Ridout's theorem therefore gives, for any $\varepsilon > 0$,

$$\left|2^{r/\ell} - \frac{p}{s \cdot 2^j}\right| > (s \cdot 2^j)^{-1-\varepsilon}$$

for all but finitely many such fractions. The proof of Theorem 2 runs with this inequality in place of an effective measure, and with $\mu = 1 + \varepsilon < \varphi$. Each exceptional fraction has a fixed nonzero distance from $2^{r/\ell}$. For an element x_i this fixes x_i and its relative error, which Lemma 11 allows only for bounded q, and for the window it is incompatible with $|\delta - 1/\ell| \to 0$. □

Theorem 3 fails for $\ell = 1$. Evidently, $q = 2^e - 1$ gives the cycle $1 \to 1$ for every $e \geq 2$. Mathematically, for $\ell \geq 2$ an odd element other than 1 puts a factor strictly between q and $q + 1$, so ℓ does not divide h and $2^{r/\ell}$ is irrational. Ridout's theorem then applies. For $\ell = 1$ that product is empty, so the same identity becomes $2^h = q + 1$, and the three conclusions of Lemma 10 fail. The root is the integer 2^h, and Ridout's theorem does not apply.

Cycles that need not visit 1 are treated in the companion paper [9]. The same finiteness, for a cycle that need not visit 1, is Theorem 2 of that paper.

8 Other results

The following were found along the way. They are not needed for Theorems 1, 2 and 3, and Corollaries 12 to 14; full statements and proofs are in the appendices indicated, and Appendix F has notes on the literature and the history of the work.

- **Cycles of length 1 and 2 through 1** (Appendix A). The case of length 2, only $q = 5$, follows from the theorem of Beukers (1980/81) on $x^2 + D = 2^n$.

- **The order of 2** (Appendix B). $k_3 = d$ or $2d$, where d is the order of 2 modulo q; in particular $d < 3 \log_2 q$, and no Wieferich prime q has a cycle of length 3 through 1, the same mechanism Franco and Pomerance use for Crandall's question of when orbits avoid 1.

- **Finiteness for length 3 via Ridout alone** (Appendix B). Lemmas 5 and 8 bound the window by about $q^{-1/2}$, and Ridout's theorem then gives finiteness through 1. This is the case $\ell = 3$ of Theorem 3. The companion paper gives the same range of lengths for cycles that need not visit 1, using the general form of Ridout's theorem.

- **Elementary special cases** (Appendix B). $k_1 = k_2 = 1$ is impossible via the integral points of $Y^2 = X^3 - 4X^2 + 12X + 16$ ($q = 41$ is a near miss), and $k_1 = 1$, $k_2 = 2$ via the factorization $(q + 2)(q^2 - q + 4)$.

- **Bounds on k_1** (Appendix B). Bauer and Bennett's inequalities give $q^{1/3}/2 < 2^{k_1} < 2.01(q+1)^{2/5}$, so each k_1 allows only q in an explicit finite range, and $k_1 = 1$ is impossible.

- **The cases of Lemma 4** (Appendix B). The cases $t > k_1$ and $t = k_1$ are impossible, and the case $t < k_1$ is tightly constrained.

- **Small elements and an earlier bound on y** (Appendix B). In fact $x \geq 5$ and $y \geq 7$. Before Lemma 6, the best result was $y \geq x$ or $y > (3q)^{1/3}$; this is the origin of the threshold $q \geq 71$ in earlier versions of the window lemma.

- **Work on cycles of length 4 through 1** (Appendix C). The analogues of Lemmas 1 and 5: $k_4 \geq k_1 + 3$, $k_4 \geq 3k_1 - 1$, and $x > \sqrt{q/2}$. The same lower bounds for y and z were left open in two configurations; they are the missing inputs for Proposition D.4. The companion paper gives ineffective finiteness for an arbitrary cycle of length 4.

- **Work on any cycle length through 1** (Appendix D). The analogues of Lemmas 1 to 3: $k_\ell \geq k_1 + 3$ and $x_{\ell-1} \equiv 1 \pmod{2^{k_1}}$, the cycle equation $2^h = q^\ell + \sum_{j=0}^{\ell-1} 2^{S_j} q^{\ell-1-j}$ with $S_0 = 0$, and $q^\ell < 2^h < (q+1)^\ell$. Before Theorem 3, finiteness for ℓ odd steps through 1 was reduced to a power of q for every element other than 1. Theorem 3 makes that reduction unnecessary.

- **Computations** (Appendix E). Searches over h for cycles through 1 with $\ell = 3$ to 20, verification of the Bauer–Bennett inputs, and the digit scan ruling out $q < 2^{1,071,999,953}$. Searches for an arbitrary cycle, and at a fixed q, are in the companion paper.

9 Open problems

1. **Length 4 with h odd.** The window and the near misses for x_1 and x_3 involve $2^{1/4}$ and $2^{3/4}$. Theorem 2 would finish the case given an effective exponent below φ for these numbers with power-of-2 denominators. The Bauer–Bennett construction needs a fourth power exceptionally close to a power of 2; among odd $a \leq 3 \times 10^6$ the best is $19^4 = 2^{17} - 751$, of quality $\log_2(2^{17})/\log_2 751 \approx 1.78$, against 4.4 for the cube case ($5^3 + 3 = 2^7$) and 5.3 for $\sqrt{2}$ ($181^2 + 7 = 2^{15}$). A different construction is likely needed.

2. **Effective results for larger ℓ.** Theorem 2 turns any such exponent for the roots $2^{r/\ell}$ into a complete answer for that ℓ. Santos's conjecture (2020) in full would need this for every ℓ, or a bound uniform in ℓ.

3. **Novelty.** Franco's 1990 thesis on Diophantine approximation and the $qx + 1$ problem could not be consulted and should be checked for overlap, including with Theorems 1, 2 and 3.

A Cycles of length 1 and 2 through 1

Proposition A.1 (length 1). The only positive odd solution of a cycle of length 1 is $x_0 = 1$ and $q = 2^h - 1$. In particular the cycle through 1 is $1 \to 1$ precisely when $q = 2^e - 1$ (Proposition 15 of Santos (2020)).

Proof. A cycle of length 1 is $C(x_0) = x_0$, so $x_0(2^h - q) = 1$. Both factors are positive integers, so $x_0 = 1$ and $2^h - q = 1$. Thus $q = 2^h - 1$. Conversely, if $q = 2^e - 1$, then $q \cdot 1 + 1 = 2^e$, so $C(1) = 1$. □

Proposition A.2 (length 2). The only odd q with 1 on a cycle of C of length 2 is $q = 5$.

Proof. The cycle gives $q(q+1) = 2^{k_1}(2^{k_2} - 1)$. Multiplying by 4 and adding 1,

$$(2q+1)^2 + (2^{k_1+2} - 1) = 2^{k_1+k_2+2}.$$

This is $X^2 + D = 2^n$ with $X = 2q+1$ and $D = 2^{k_1+2} - 1$. By the theorem of Beukers (1980/81), stated as Theorem 1.8 of Bauer and Bennett (2002), for $D \neq 7$, 23 and D not of the form $2^k - 1$ there is at most one solution, while for $D = 2^k - 1$ with $k \geq 4$ there are exactly two, $X = 1$ and $X = 2^{k-1} - 1$. These give $q = 0$ and $q = 2^{k_1} - 1$, the one-step cycle traversed twice, so $k_1 \geq 2$ gives nothing new. For $k_1 = 1$, $D = 7$ is the Ramanujan–Nagell equation with $X \in \{1, 3, 5, 11, 181\}$, and only $X = 11$ gives an odd q other than 1: $q = 5$, with $1 \to 3 \to 1$. □

As a check, the argument of Lemma B.2 below gives $2^{k_2} = qx + 1 < q^2 < 2^{2d}$ with d the order of 2 modulo q, so $k_2 = d$, and the cycle condition reads $q(q+1) = 2^{k_1}(2^d - 1)$; for $q = 5$, $d = 4$ and $5 \cdot 6 = 2 \cdot 15$.

B Earlier results on cycles of length 3 through 1

These results were found before Theorem 1 and are not needed for it. Notation is as in Sections 3 to 5. Bauer and Bennett's headline inequality, $|x^3 - 2^n| \geq x^{4/3}$ unless $x^3 = 2^n$ or $(x, n) = (5, 7)$, and their Corollary 4, $|x^3 - 27 \cdot 2^n| \geq 3^{5/3} x^{4/3}$ unless $x^3 = 27 \cdot 2^n$ or $x \in \{4, 5, 8, 15, 19, 38, 121\}$, are used throughout; both were also checked numerically for $n < 200$ and $n < 70$ respectively.

Factorization and the order of 2.

Lemma B.1 (factorization). With $q = 2^{k_1} x - 1$,

$$(2^{k_1} x - 1)(2^{k_1} x^2 - x + 1) = 2^{k_2}(2^{k_3} - 1).$$

Proof. The factors are q and $qx + 1$, and $q(qx+1) = q \cdot 2^{k_2} y = 2^{k_2}(2^{k_3} - 1)$. □

Completing the square in $qx + 1 = 2^{k_2} y$ gives $(2q+1)^2 + (2^{k_1+2} - 1) = 2^{k_1+k_2+2} \cdot y$, so the three-step case is the equation of Proposition A.2 with an odd cofactor y, tied to q by $qy + 1 = 2^{k_3}$.

Lemma B.2 (order of 2). $k_3 = d$ or $k_3 = 2d$, where d is the order of 2 modulo q. In particular $d < 3 \log_2 q$.

Proof. $2^{k_3} = qy + 1 \equiv 1 \pmod{q}$, so d divides k_3. Using $2^{k_1} \geq 2$ and $2^{k_2} \geq 2$, $y \leq (q^2 + q + 2)/4$, so $2^{k_3} \leq q(q^2 + q + 2)/4 + 1 < q^3$. Also $2^d \geq q + 1 > q$, so $q^3 < 2^{3d}$, and hence $k_3 < 3d$. □

Corollary B.3 (Wieferich primes). Let $F = (2^d-1)/q$ and $k_3 = md$. Then $mF \equiv 2^{-k_2} \pmod q$, so $\gcd(F, q) = 1$. In particular no Wieferich prime q ($q^2 \mid 2^{q-1} - 1$) has a cycle of length 3 through 1.

Proof. Since $2^d \equiv 1 \pmod q$, $y = (2^{md}-1)/q = F(1+2^d+\cdots+2^{(m-1)d}) \equiv mF \pmod q$. Reducing $2^{k_2}y = qx + 1 \bmod q$ gives $2^{k_2} \cdot mF \equiv 1$. For a Wieferich prime, the lifting-the-exponent lemma shows that q^2 divides $2^d - 1$, since q^2 divides $2^{q-1} - 1$ and q does not divide $(q-1)/d$; so $q \mid F$. □

This is the mechanism Franco and Pomerance use for Crandall's question of when orbits avoid 1. Together with Lemma 3, Lemma B.2 determines everything from q: $k_1 = v_2(q+1)$, h is the unique exponent with $q^3 < 2^h < (q+1)^3$, $k_3 \in \{d, 2d\}$ and $k_2 = h - k_1 - k_3$.

Finiteness via Ridout's theorem.

Proposition B.4. Only finitely many odd q have a cycle of length 3 through 1.

Proof. By Lemmas 8 and 5, $\delta - 1/3 < 2^{k_1+1}/(3q)$ with $2^{k_1} = (q+1)/x < (q+1)\sqrt{2/q}$, so $\delta - 1/3 < 0.96\, q^{-1/2}$ for $q \geq 71$. With $h = 3m + r$, $\beta = 2^{r/3}$, $P = 3q + 1$ and $Q = 3 \cdot 2^m$ this gives $0 < \beta - P/Q < 2.9\, q^{-1/2}/Q$, and since $Q < 2.39(q+1)$ we get $|\beta - P/Q| < 4.6\, Q^{-3/2} < Q^{-5/4}$ for $Q > 450$. The reduced fraction still has a denominator built from 2 and 3, so Ridout's theorem with $S = \{2, 3\}$ and $\varepsilon = 1/4$ allows only finitely many such fractions; infinitely many q would give infinitely many distinct ones, since the errors tend to 0 and β is irrational. □

This is the case $\ell = 3$ of Theorem 3, and Theorem 1 supersedes it. The exponent $3/2$ comes from the lower bounds $x > \sqrt{q/2}$ and $y > x$; with only $x, y \geq 3$ the error is of order $1/Q$ and Ridout's theorem gives nothing. The result is ineffective, which is why Theorem 1 needs Theorem (Bauer–Bennett (2018)) instead.

Elementary special cases.

Proposition B.5 ($k_1 = k_2 = 1$). No cycle of length 3 through 1 has $k_1 = k_2 = 1$. Hence $q \not\equiv 1 \pmod 8$.

Proof. Lemma 2 gives $2^h = q^3 + q^2 + 2q + 4 = (q+1)(q^2+2) + 2$. With $q + 1 = 2x$ this becomes

$$4x^3 - 4x^2 + 3x + 1 = 2^n, \qquad n = h - 1,$$

and $q^2 + 2 = 4x^2 - 4x + 3$ divides $2^n - 1$. Since q is odd, $q^2 + 2 \equiv 3 \pmod 8$. Every prime p dividing $q^2 + 2$ has -2 as a square mod p, so $p \equiv 1$ or $3 \pmod 8$, and at least one is $\equiv 3 \pmod 8$. For that p, 2 is not a square mod p, so the order of 2 modulo p is even and divides n. So n is even, and $(X, Y) = (4x,\ 4 \cdot 2^{n/2})$ is an integral point on

$$E\colon Y^2 = X^3 - 4X^2 + 12X + 16.$$

Using Sage (passagemath) with proofs enabled, E has rank 2 and trivial torsion, generated by $(0, 4)$ and $(1, 5)$ with saturation index 1, and its integral points are $X \in \{0, 1, 4, 9, 84\}$, computed by the

elliptic-logarithm method. $X = 4$ gives $q = 1$; $X = 84$ gives $x = 21$ but $Y = 752 = 4 \cdot 188$, and 188 is not a power of 2; the others do not have the form $4x$ with x odd and at least 3. For the congruence: if $q \equiv 1 \pmod 8$, then $k_1 = 1$ and $x \equiv 1 \pmod 4$, so $t > k_1$ in Lemma 4, and its proof forces $k_2 = k_1 = 1$. □

The point $X = 84$ is a near miss: for $q = 41$, $q^3 + q^2 + 2q + 4$ is twice a perfect square, and the cycle fails only at the last step, since $41 \cdot 431 + 1 = 17{,}672 = 8 \cdot 47^2$.

Proposition B.6 ($k_1 = 1$, $k_2 = 2$). No cycle of length 3 through 1 has $k_1 = 1$ and $k_2 = 2$. Hence $q \not\equiv 13 \pmod{16}$.

Proof. Lemma 2 gives $2^h = q^3 + q^2 + 2q + 8 = (q + 2)(q^2 - q + 4)$, and $q + 2$ is an odd number at least 5. For $k_1 = 1$, $k_2 = 2$ holds exactly when $x \equiv 7 \pmod 8$, that is $q \equiv 13 \pmod{16}$. □

Both are superseded by Corollary B.8, which rules out $k_1 = 1$ altogether.

Effective bounds on k_1.

Proposition B.7. In any cycle of length 3 through 1, $2^{k_1} > q^{1/3}/2$.

Proof. Apply Bauer and Bennett's Corollary 4 with $x = 3q+1$ and $n = h$. The exceptional values of x give $q \in \{1, 6, 40\}$, which are excluded, and $N > 0$ by Lemma 8, so $N \geq 3^{5/3}(3q + 1)^{4/3} > 27\, q^{4/3}$. By Lemma 8, $N < 54 \cdot 2^{k_1} q$. Comparing, $54 \cdot 2^{k_1} q > 27\, q^{4/3}$. □

Corollary B.8. (a) $k_1 = 1$ is impossible: it would force $q < 64$, and the direct search excludes every $q < 2{,}000{,}000$; so every possible q is $\equiv 3 \pmod 4$. (b) Together with Lemma 5, $2^{2k_1-1} - 1 < q < 2^{3k_1+3}$, so each k_1 allows only an explicit finite range of q. (c) Since the digit scan (Appendix E) excludes $q < 2^{1{,}071{,}999{,}953}$, any remaining q would need $k_1 \geq 357{,}333{,}317$, so $q \equiv -1 \pmod{2^{357{,}333{,}317}}$.

Theorem 1 rules out these q altogether.

Proposition B.9 (upper bound on k_1). $2^{k_1} \leq 3.2\, x^{2/3}$, and hence $2^{k_1} < 2.01(q + 1)^{2/5}$.

Proof. By Lemma 9, $\left|x^3 - 2^f\right| \leq 3.2\, x^2/2^{k_1}$ and $\left|x - 2^{f/3}\right| \leq 1.6/2^{k_1} < 1$, so $x^3 \neq 2^f$. Bauer and Bennett's headline inequality gives $\left|x^3 - 2^f\right| \geq x^{4/3}$, and therefore $2^{3k_1} x^{4/3} \leq 3.2 \cdot 2^{2k_1} x^2$. The exception $x = 5$ forces $q \leq 39$ by Lemma 5. □

Together with Proposition B.7, $q^{1/3}/2 < 2^{k_1} < 2.01(q + 1)^{2/5}$.

The three cases of Lemma 4.

Proposition B.10 ($t > k_1$ is impossible). No cycle of length 3 through 1 has $t > k_1$.

Proof. Lemma 4 and its proof give $t = 2k_1$ and $k_2 = k_1$, so $x = 1+2^{2k_1}s$ with s odd. Proposition B.7 gives $q < 8 \cdot 2^{3k_1}$, so $x < 8 \cdot 2^{2k_1}$ and $s \in \{1, 3, 5, 7\}$. Then $q = s \cdot 2^{3k_1} + 2^{k_1} - 1$, and the cycle equation reads $2^h = q^3 + q^2 + 2^{k_1}q + 2^{2k_1}$. Dividing by 2^{9k_1},

$$s^3 < \frac{2^h}{2^{9k_1}} < \left(s + \frac{1}{2^{2k_1}}\right)^3 \left(1 + \frac{2}{q}\right),$$

and for $2^{k_1} \geq 8$ the right-hand side is below the next power of 2 after s^3 (2, 32, 128, 512), so $2^h/2^{9k_1}$ lies strictly between consecutive powers of 2. For $2^{k_1} \leq 4$, $q \leq 451$, which the direct search covers. □

(Also checked in exact arithmetic for $k_1 < 400$.)

Proposition B.11 ($t = k_1$ is impossible). No cycle of length 3 through 1 has $t = k_1$.

Proof. Here $x = 1 + 2^{k_1}s$ with s odd, so $q = 2^{2k_1}s + 2^{k_1} - 1$. Let $g = h - 6k_1$. Expanding the cycle equation,

$$0 < 2^g - s^3 = \frac{R}{2^{6k_1}} \leq \frac{6s^2}{2^{k_1}}, \qquad R = 3{\cdot}2^{4k_1}s^2(2^{k_1}-1)+3{\cdot}2^{2k_1}s(2^{k_1}-1)^2+(2^{k_1}-1)^3+q^2+2^{k_1}q+2^{k_1+k_2},$$

using $q < 2{\cdot}2^{2k_1}s$ and $2^{k_2} < q$ ($2^{k_1} \geq 4$). Bauer and Bennett's headline inequality gives $2^g - s^3 \geq s^{4/3}$, so $2^{k_1} \leq 6\,s^{2/3}$, while Proposition B.7 gives $8 \cdot 2^{3k_1} > q > 2^{2k_1}s$, so $2^{k_1} > s/8$. Hence $s < 48^3$ and $2^{k_1} \leq 13{,}824$, so $q < 2 \cdot 2^{2k_1}s < 5 \times 10^{13}$, which the computations cover. The exception $s = 5$ gives $q < 4 \times 10^6$ through Proposition B.9. □

Proposition B.12 (a constraint when $t < k_1$). Suppose $t < k_1$, so that $k_2 = t$, and write $x = 2^{k_1}u - (2^{k_2} - 1)$ with u odd. Then $2^{k_1}/2^{k_2} \leq 3u^{2/3}$.

Proof. Put $\delta' = 2^{k_1+k_2} - 2^{k_1} + 1$, so $q = 2^{2k_1}u - \delta'$, and $g = h - 6k_1$. Expanding,

$$0 < u^3 - 2^g = \frac{3 \cdot 2^{4k_1}u^2\delta' - 3 \cdot 2^{2k_1}u\delta'^2 + \delta'^3 - q^2 - 2^{k_1}q - 2^{k_1+k_2}}{2^{6k_1}} \leq \frac{3u^2 \cdot 2^{k_2}}{2^{k_1}},$$

since the terms after the first are negative in total and $\delta' < 2^{k_1+k_2}$. Bauer and Bennett's headline inequality gives $u^3 - 2^g \geq u^{4/3}$. The exceptions are $u = 1$, where $x \leq 2^{k_1}$ and Proposition B.9 forces $2^{k_1} \leq 32$, and $u = 5$, where it forces $q < 4 \times 10^6$. □

The positivity and upper bounds in Propositions B.11 and B.12 were also checked in exact arithmetic on hundreds of thousands of random parameter choices. Propositions B.10 and B.11 leave only the case $t < k_1$; Theorem 1 makes the case analysis unnecessary.

Small elements and an earlier bound on y.

Remark (small elements). The elements x and y are odd and different from 1, so $x, y \geq 3$, and Lemma 7 gives $y \geq 5$. If $x = 3$, Lemma 5 gives $2^{k_1} \leq 2(x-1) = 4$, so $q = 3 \cdot 2^{k_1} - 1 \in \{5, 11\}$, and neither has a cycle of length 3 through 1. Hence $x \geq 5$ and $y \geq 7$. None of the proofs in the main text need more than $x \geq 3$ and $y > x$.

Before Lemma 6 was found, the case $y < x$ could not be excluded, and the following weaker bound was used instead. Lemma 7 shows that $y < x$ never happens, so the second alternative below is vacuous.

Proposition B.13. In a cycle of length 3 through 1, either $y \geq x$ or

$$y > \left(\frac{3q^2}{q+1}\right)^{1/3} \approx (3q)^{1/3}.$$

Moreover, in the second case $v_2(y-1) = k_1$ exactly.

Proof. Multiplying $2^{k_2} y = qx + 1$ by $2^{k_1} y^2$ and using $qy = 2^{k_3} - 1$ and $q + 1 = 2^{k_1} x$ gives the exact identity

$$2^{2k_3} + (y-2)\, 2^{k_3} + C = 2^{k_1+k_2} y^3, \qquad C = 2^{k_1} y^2 - y + 1.$$

If 2^{k_3} divides C, then $C \geq 2^{k_3} = qy + 1$, which simplifies to $2^{k_1} y \geq q + 1$, that is $y \geq x$. Otherwise the first two terms are divisible by a higher power of 2 than C, so $v_2(C) = k_1 + k_2$ and $2^{k_1+k_2} \leq C < 2^{k_1} y^2$. Since the left side exceeds $2^{2k_3} > q^2 y^2$, this gives $q^2 y^2 < 2^{k_1+k_2} y^3 < 2^{k_1} y^5$, so $q^2 < 2^{k_1} y^3 \leq (q+1) y^3/3$. For the last claim, $y \equiv 1 \pmod{2^{k_1}}$ by Lemma 1; if $v_2(y-1) > k_1$, then $v_2(C) = k_1$, forcing $k_2 = 0$, which is impossible. □

For $q \geq 71$ the cube root is the smaller of the two bounds, so this gave $y > 1.436\, q^{1/3}$ unconditionally, and for $q >$ 2,000,000 it gave $y \geq 183$. The identity was also checked by computer algebra: substituting the cycle relations into both sides and expanding gives the same expression.

C Work on cycles of length 4 through 1

The first and last steps. Write a cycle of length 4 through 1 as $1 \to x \to y \to z \to 1$, with

$$q + 1 = 2^{k_1} x, \qquad qx + 1 = 2^{k_2} y, \qquad qy + 1 = 2^{k_3} z, \qquad qz + 1 = 2^{k_4}.$$

The next three lemmas are the four-step forms of Lemma 1 and Lemma 5, together with the gain from a larger x in the last step. They constrain k_4 and x, not y or z.

Lemma C.1. $k_4 \geq k_1 + 3$, and $z \equiv 1 \pmod{2^{k_1}}$.

Proof. The odd elements other than 1 are at least 3, so $x \geq 3$ and $z \geq 3$. The first step is $q+1 = 2^{k_1} x$, and therefore $q + 1 \geq 3 \cdot 2^{k_1}$, that is $q \geq 3 \cdot 2^{k_1} - 1$. The last step is $2^{k_4} = qz + 1$, so

$$2^{k_4} \geq (3 \cdot 2^{k_1} - 1) \cdot 3 + 1 = 9 \cdot 2^{k_1} - 2.$$

Since $2^{k_1} \geq 2$, we have $9 \cdot 2^{k_1} - 2 \geq 8 \cdot 2^{k_1} = 2^{k_1+3}$, and hence $k_4 \geq k_1 + 3$. From $q + 1 = 2^{k_1} x$ we also have $q \equiv -1 \pmod{2^{k_1}}$. Thus $qz + 1 \equiv 1 - z \pmod{2^{k_1}}$. But $qz + 1 = 2^{k_4}$ is divisible by 2^{k_1}, because $k_4 \geq k_1$, so $1 - z \equiv 0 \pmod{2^{k_1}}$, that is $z \equiv 1 \pmod{2^{k_1}}$. □

Lemma C.2. $2^{k_1} \le 2(x-1)$ and $x > \sqrt{q/2}$.

Proof. By Lemma C.1, $z \equiv 1 \pmod{2^{k_1}}$, so $qy + 1 = 2^{k_3} z$ gives $y \equiv 1 - 2^{k_3} \pmod{2^{k_1}}$. Let $t = v_2(x-1)$ and $s = (x-1)/2^t$. If $t \ge k_1$, then $2^{k_1} \le x - 1$. Otherwise $k_2 = t$ and $y = 2^{k_1 - t} x^2 - s \equiv 2^{k_1 - t} - s \pmod{2^{k_1}}$, since $x^2 \equiv 1 \pmod{2^t}$. So $M = 2^{k_1 - t} + 2^{k_3} - (s+1)$ is a multiple of 2^{k_1}. If $M = 0$, then $2^{k_1 - t} < s + 1$, so $2^{k_1} < 2^t(s+1) \le 2(x-1)$. If $M \le -2^{k_1}$, then $s + 1 > 2^{k_1}$, so $2^{k_1} \le x$. If $M \ge 2^{k_1}$, then $2^{k_3} > 2^{k_1 - 1}$, so $k_3 \ge k_1$ and $2^{k_1 - t} \equiv s + 1 \pmod{2^{k_1}}$, giving $2^{k_1 - t} = s + 1$ or $s + 1 \ge 2^{k_1}$; either way $2^{k_1} \le 2(x-1)$. The first step is $q + 1 = 2^{k_1} x$, so

$$q + 1 \le 2x(x-1) = 2x^2 - 2x.$$

Hence $q < 2x^2$, that is $x > \sqrt{q/2}$. □

Lemma C.3. $k_4 \ge 2k_1 + 1$ and $k_4 \ge 3k_1 - 1$.

Proof. By Lemma C.2, $x \ge 2^{k_1 - 1} + 1$. Since $z \ge 3$, the last step gives $2^{k_4} = qz + 1 \ge 3q + 1$. The first step is $q = 2^{k_1} x - 1$, so $3q + 1 = 3 \cdot 2^{k_1} x - 2$, and therefore

$$2^{k_4} \ge 3 \cdot 2^{2k_1 - 1} + 3 \cdot 2^{k_1} - 2 > 2^{2k_1},$$

so $k_4 \ge 2k_1 + 1$.

By Lemma C.1, $z \equiv 1 \pmod{2^{k_1}}$ and $z \ne 1$, so $z \ge 1 + 2^{k_1}$. Then $2^{k_4} = qz + 1 \ge (2^{k_1} x - 1)(2^{k_1} + 1) + 1 = 2^{k_1}(x(2^{k_1} + 1) - 1)$. Substituting $x \ge 2^{k_1 - 1} + 1$ yields

$$x(2^{k_1} + 1) - 1 \ge (2^{k_1 - 1} + 1)(2^{k_1} + 1) - 1 = 2^{2k_1 - 1} + 2^{k_1 - 1} + 2^{k_1},$$

hence

$$2^{k_4} \ge 2^{3k_1 - 1} + 2^{2k_1 - 1} + 2^{2k_1} = 2^{3k_1 - 1} + 3 \cdot 2^{2k_1 - 1} > 2^{3k_1 - 1},$$

so $k_4 \ge 3k_1 - 1$. □

The middle elements.

Lemma C.4. If $k_2 \le k_3$, then either $y > q$, or $y > x$ and $y^2 > qx$. In particular $y \gg q^{3/4}$. If $k_3 \le k_4$, then either $z > q$, or $z > y$ and $z^2 > qy$.

Proof. Apply Lemma 6 to $x \to y \to z$, so $J = k_2$ and $K = k_3$. If $k_2 \le k_3$ and $y < x$, then $y > q$; if $y > x$, then $y^2 > qx$. In the first case $y \gg q^{3/4}$. In the second, Lemma C.2 gives $x > \sqrt{q/2}$, so $y^2 > q^{3/2}/\sqrt{2}$ and again $y \gg q^{3/4}$. Apply Lemma 6 to $y \to z \to 1$, so $J = k_3$ and $K = k_4$. If $k_3 \le k_4$ and $z < y$, then $z > q$; if $z > y$, then $z^2 > qy$. □

Lemma C.4 bounds y only when $k_2 \le k_3$, and z only when $k_3 \le k_4$. It leaves open $k_3 < k_2$ with $y < x$, where y could be small, and $k_4 < k_3$, where z could be small. Those two lower bounds are the missing inputs for Proposition D.4. Theorem 2 of the companion paper does not need them.

D Work on any cycle length through 1

The structure of a cycle. The next three lemmas are Lemmas 1 to 3 for an arbitrary length.

Lemma D.1. Let $\ell \geq 2$. On a cycle of length ℓ through 1, $k_\ell \geq k_1 + 3$ and $x_{\ell-1} \equiv 1 \pmod{2^{k_1}}$.

Proof. Each of x_1 and $x_{\ell-1}$ is an odd element other than 1, so $x_1 \geq 3$ and $x_{\ell-1} \geq 3$. The first step is $q + 1 = 2^{k_1} x_1$, so $q + 1 \geq 3 \cdot 2^{k_1}$ and $q \geq 3 \cdot 2^{k_1} - 1$. The last step is $2^{k_\ell} = q x_{\ell-1} + 1$. Since $q \geq 3 \cdot 2^{k_1} - 1$ and $x_{\ell-1} \geq 3$,

$$2^{k_\ell} = q x_{\ell-1} + 1 \geq (3 \cdot 2^{k_1} - 1) \cdot 3 + 1 = 9 \cdot 2^{k_1} - 2.$$

Since q is odd, $q + 1$ is even, so $k_1 \geq 1$ and $2^{k_1} \geq 2$. Then $9 \cdot 2^{k_1} - 2 - 8 \cdot 2^{k_1} = 2^{k_1} - 2 \geq 0$, and therefore $9 \cdot 2^{k_1} - 2 \geq 8 \cdot 2^{k_1} = 2^{k_1+3}$. Hence $2^{k_\ell} \geq 2^{k_1+3}$, so $k_\ell \geq k_1 + 3$.

From $q + 1 = 2^{k_1} x_1$, one has $q \equiv -1 \pmod{2^{k_1}}$. Since $k_\ell \geq k_1 + 3$, one has $k_\ell \geq k_1$, so $2^{k_\ell} \equiv 0 \pmod{2^{k_1}}$. The last step is $q x_{\ell-1} + 1 = 2^{k_\ell}$, so

$$1 - x_{\ell-1} \equiv q x_{\ell-1} + 1 \equiv 2^{k_\ell} \equiv 0 \pmod{2^{k_1}}.$$

Hence $x_{\ell-1} \equiv 1 \pmod{2^{k_1}}$. □

Lemma 1 is the case $\ell = 3$, and Lemma C.1 is the case $\ell = 4$.

Lemma D.2 (cycle equation). For a cycle of length ℓ through 1, set $S_0 = 0$. Then

$$2^h = q^\ell + \sum_{j=0}^{\ell-1} 2^{S_j} q^{\ell-1-j}.$$

Proof. For $1 \leq m \leq \ell$,

$$2^{S_m} x_m = q^m + \sum_{j=0}^{m-1} 2^{S_j} q^{m-1-j}.$$

The case $m = 1$ is the first step $2^{k_1} x_1 = q + 1$. Suppose $m \geq 2$ and

$$2^{S_{m-1}} x_{m-1} = q^{m-1} + \sum_{j=0}^{m-2} 2^{S_j} q^{m-2-j}.$$

The step from x_{m-1} is $2^{k_m} x_m = q x_{m-1} + 1$, and $S_m = S_{m-1} + k_m$, so

$$2^{S_m} x_m = 2^{S_{m-1}} (q x_{m-1} + 1) = q \cdot 2^{S_{m-1}} x_{m-1} + 2^{S_{m-1}}.$$

Substituting the equation for $m - 1$ gives

$$2^{S_m} x_m = q \left(q^{m-1} + \sum_{j=0}^{m-2} 2^{S_j} q^{m-2-j} \right) + 2^{S_{m-1}} = q^m + \sum_{j=0}^{m-1} 2^{S_j} q^{m-1-j}.$$

At $m = \ell$, one has $x_\ell = 1$ and $S_\ell = h$, so the left side is 2^h and the right side is $q^\ell + \sum_{j=0}^{\ell-1} 2^{S_j} q^{\ell-1-j}$. □

Lemma 2 is the case $\ell = 3$.

Lemma D.3. For a cycle of length ℓ through 1 with $\ell \geq 2$,

$$q^\ell < 2^h < (q+1)^\ell.$$

Hence ℓ does not divide h and $q = \lfloor 2^{h/\ell} \rfloor$.

Proof. Every term $2^{S_j} q^{\ell-1-j}$ in Lemma D.2 is positive, so $2^h > q^\ell$. Multiplying $2^{k_i} = (qx_{i-1}+1)/x_i$ over $1 \leq i \leq \ell$ gives

$$2^h = \prod_{i=1}^{\ell} \frac{qx_{i-1}+1}{x_i} = \frac{(q+1)\prod_{i=1}^{\ell-1}(qx_i+1)}{(\prod_{i=1}^{\ell-1} x_i)\, x_\ell} = (q+1)\prod_{i=1}^{\ell-1}\left(q + \frac{1}{x_i}\right),$$

since $x_\ell = 1$. Each x_i with $1 \leq i \leq \ell - 1$ is an odd element other than 1, so $x_i \geq 3$ and $q + 1/x_i \leq q + 1/3 < q + 1$. Therefore $2^h < (q+1)^\ell$. A power of 2 strictly between consecutive ℓ-th powers is not an ℓ-th power, so $\ell \nmid h$. Thus $q < 2^{h/\ell} < q + 1$, and $q = \lfloor 2^{h/\ell} \rfloor$. □

Lemma 3 is the case $\ell = 3$. Lemma 10 includes these bounds.

Reduction to lower bounds. Before Theorem 3, finiteness for ℓ odd steps through 1 was reduced to a power of q for every element other than 1.

Proposition D.4 (reduction to lower bounds). Fix ℓ. Suppose there are fixed real numbers $\theta > 0$ and $c > 0$ such that, on every cycle of length ℓ through 1, every element other than 1 is at least $c \cdot q^\theta$. Then only finitely many q have a cycle of length ℓ through 1.

Proof. As for Proposition B.4, with $h = \ell m + r$, $\beta = 2^{r/\ell}$, $P = \ell q + 1$ and $Q = \ell \cdot 2^m$: Lemma 10 gives $|\beta - P/Q| \ll Q^{-1-\theta}$, and Ridout's theorem applies with $S = \{2\} \cup \{\text{primes dividing } \ell\}$. □

This is an earlier and weaker result than Theorem 3. It is conditional on the hypothesis that every element other than 1 is at least $c \cdot q^\theta$ for fixed real numbers $\theta > 0$ and $c > 0$, whereas Theorem 3 is not.

E Computations

For a cycle of length ℓ through 1 with $\ell \geq 2$, Lemma 10 gives $q^\ell < 2^h < (q+1)^\ell$, so $\ell \nmid h$ and $q = \lfloor 2^{h/\ell} \rfloor$. Length 1 is Proposition A.1, and is not searched. A search of every $h \leq H$ with $\ell \nmid h$ therefore excludes every such cycle with $q < 2^{\lfloor H/\ell \rfloor}$: it would have $2^h < (q+1)^\ell \leq 2^{\ell \lfloor H/\ell \rfloor} \leq 2^H$, so $h \leq H$. Searches for a cycle that need not visit 1, and searches at a fixed q, are in the companion paper [9].

Cycles through 1. For each ℓ and each $h \leq H$ with $\ell \nmid h$, the candidate $q = \lfloor 2^{h/\ell} \rfloor$ is tested by following the orbit of 1 for ℓ odd steps. For $\ell = 3$, every $h \leq 200{,}000$ was tested. Theorem 1 needs only $h < 66{,}003$: $q < 2^{22001}$ and Lemma 3 give $2^h < 2^{66{,}003}$. For each ℓ from 4 to 20, every $h \leq 100{,}000$ was tested. No cycle through 1 was found.

Cycles through 1

length ℓ	h at most	no cycle through 1 for q below
3	200,000	$2^{66,666}$
4	100,000	$2^{25,000}$
5	100,000	$2^{20,000}$
6	100,000	$2^{16,666}$
7	100,000	$2^{14,285}$
8	100,000	$2^{12,500}$
9	100,000	$2^{11,111}$
10	100,000	$2^{10,000}$
11	100,000	$2^{9,090}$
12	100,000	$2^{8,333}$
13	100,000	$2^{7,692}$
14	100,000	$2^{7,142}$
15	100,000	$2^{6,666}$
16	100,000	$2^{6,250}$
17	100,000	$2^{5,882}$
18	100,000	$2^{5,555}$
19	100,000	$2^{5,263}$
20	100,000	$2^{5,000}$

Checks of Lemmas 2, 4, 6, 8 and 9. The identities in these lemmas were checked by computer algebra, substituting the cycle relations and expanding both sides, or on random orbits. Lemma 4 was also checked numerically for every $k_1 \le 7$ and every odd $x < 20{,}000$. In step 3 of the proof of Theorem 1, the two bounds on a were compared for every f from 33,000 to 400,000.

Digits of $2^{1/3}$ and $2^{2/3}$. A cycle of length 3 through 1 forces a long alternating block in the binary expansion of $2^{1/3}$ or of $2^{2/3}$. The block is not present in the digits that were computed, except for $m < 100$, which the search over h already covers.

Proposition E.1 (digit test). In a cycle of length 3 through 1 with $q \ge 17$, write $h = 3m + r$ with $r \in \{1, 2\}$ and set $\beta = 2^{r/3}$. Then $q = \lfloor 2^m \beta \rfloor$. Places $m + 1$ through $m + \lfloor m/2 \rfloor - 2$ in the binary expansion of the fractional part of β alternate $0, 1, 0, 1, ...$, beginning with 0. In particular, for $m \ge 100$, places $m + 1$ through $m + 48$ alternate in the same way.

Proof. Lemma 3 gives $q = \lfloor 2^{h/3} \rfloor$. Since $h = 3m + r$, one has $2^{h/3} = 2^m \beta$, so $q = \lfloor 2^m \beta \rfloor$. The fractional part of $2^m \beta$ is δ, and its binary digits are the digits of the fractional part of β from place $m + 1$ on. Lemma 8 gives $0 < \delta - 1/3 < 2^{k_1+1}/(3q)$; the upper bound uses $2^{k_1+k_2} \le 2^{k_1} q$ from Lemma 7. The first step is $q + 1 = 2^{k_1} x$, and Lemma 5 gives $x > \sqrt{q/2}$, so $2^{k_1} < (q + 1)\sqrt{2/q}$ and

$$0 < \delta - \frac{1}{3} < \frac{2\sqrt{2}}{3} \left(1 + \frac{1}{q}\right) q^{-1/2}.$$

For $q \ge 17$ one has $1 + 1/q \le 18/17$, so $\frac{2\sqrt{2}}{3}(1 + \frac{1}{q}) \le 12\sqrt{2}/17 < 1$, because $\sqrt{2} < 17/12$. Thus $\delta - 1/3 < q^{-1/2}$. Also $\beta > 1$, so $q = \lfloor 2^m \beta \rfloor \ge 2^m$ and $q^{-1/2} \le 2^{-m/2}$. Let $k = \lfloor m/2 \rfloor$. Then $m/2 \ge k$, so $2^{-m/2} \le 2^{-k}$, and

$$\frac{1}{3} < \delta < \frac{1}{3} + 2^{-k}.$$

In binary, $1/3 = 0.010101\ldots_2$: place $2j-1$ is 0 and place $2j$ is 1. Adding a positive amount less than 2^{-k} can change place k, and the carry stops at place $k-1$, because each 1 in this expansion is preceded by a 0. Places 1 through $k-2$ of δ therefore alternate $0, 1, 0, 1, \ldots$, beginning with 0. These are places $m+1$ through $m+k-2$ of the fractional part of β. For $m \geq 100$ one has $k-2 \geq 48$, so places $m+1$ through $m+48$ alternate in the same way. □

Let $N = 1{,}072{,}000{,}000$. The first N binary places of $\beta = 2^{1/3}$ and of $\beta = 2^{2/3}$ were computed and checked by $I^3 < 2^{3N+r} < (I+1)^3$, where $I = \lfloor 2^N \beta \rfloor$. A multiprecision value was kept only when its 192 guard digits were neither all 0 nor all 1 and its leading 61 digits matched an integer cube root. Above about 2^{30} digits that value disagreed with the integer test, so N was kept below 2^{30} and the integer test is the one used here. For $m \geq 100$, Proposition E.1 requires places $m+1$ through $m+48$ to alternate. No such block occurs for any m with $m+48 \leq N$, that is $m \leq N-48$. Such an m would give $q = \lfloor 2^m \beta \rfloor < 2^{m+1} \leq 2^{N-47} = 2^{1{,}071{,}999{,}953}$, since $\beta < 2$. Every $m < 100$ has $h = 3m + r < 300$, which the search $h \leq 200{,}000$ includes. Together these rule out every cycle of length 3 through 1 with $q < 2^{1{,}071{,}999{,}953}$.

Checking Bauer and Bennett (2018). The results of this paper rely heavily on Bauer and Bennett (2018), so the steps that produce their inequality were checked. The labels below are those of the published paper. In the author's preprint the same results are Theorem 3, inequality (21), Lemma 7, Proposition 8 and Lemma 9.

Theorem (Bauer–Bennett (2018)) is the inequality in the proof of their Theorem 2.1, before the exponent is rounded. The constant 16166467234 in that inequality is the integer just above $9 \cdot 2^{28} \cdot F(\tau)/1.79$, where F is their auxiliary function: $F(\tau) = 11.97804\ldots$ at $\tau = 0.20305\ldots$, and the product equals $16166467233.3\ldots$. The denominator of their inequality (5.3) contributes $1/(1.26 \cdot 2^{14/3}) = 2^{-5.00009\ldots}$, and their displayed intermediate step omits this factor. The factor 2^{-5} in Theorem (Bauer–Bennett (2018)) is a weakening of the contour bound for $|P_{n_1,\delta}|$, which has constant $1.2528\ldots$ in place of 1.26: then $1.2528\ldots \cdot 2^{14/3}$ has base-2 logarithm $4.9918 < 5$. The sentence before the proof of Theorem 1 records the same comparison, and the factor is absorbed in their final bound $2^{-1.62k}$.

Their Lemma 4.1 was checked at 3000-digit precision for every parameter n_1 from 2 to 60 and both values of each of δ and r. The inequality uses $n_1 \geq 497$, and Lemma 4.1 is uniform in n_1. With

$$f_{\delta,r}(v) = \frac{v^{4+r/3-\delta}(1-v)^{1-r/3}}{\left(1 - 3(1-v)/128\right)^{5-\delta}},$$

one has $\max f_{\delta,r} = 0.31999\ldots < 1/3$, and $F(\tau) < 12$.

Their Proposition 5.1 states that a product Π, built from their Padé approximants, satisfies $\Pi > 1.8^{n_1}$ for every integer $n_1 \geq 497$. It was recomputed as follows.

- Their Lemma 6.1 lists the primes that divide every Padé coefficient. For every $n_1 \leq 160$ and both values of each of δ and r, the 5,978 primes that meet its hypotheses were checked, with no failure.

- A lower bound $L(r, n_1)$ for $\log \Pi$ was computed exactly for every $497 \leq n_1 < 10^6$ and both values of r. It lower-bounds $\log \Pi$ through Lemma 6.1. The cutoff $\lfloor \sqrt{n_1}/12 \rfloor - 2$ in

that application is negative for $n_1 < 576$, so on $n_1 \le 830$ the sum includes primes beyond the cutoff, each still required to satisfy $p > \max\{\sqrt{3n_2+2},\, 5\}$. It reproduces their values $\exp(L(2,3319)/3319) = 1.897735\ldots$ and $\exp(L(1,10000)/10000) = 2.006523\ldots$. The bound exceeds $\log 1.8$ except for 43 pairs (n_1, r), all with $n_1 \le 830$. For those pairs Π was computed exactly for both values of δ, and the minimum of Π^{1/n_1} is 1.80767, at $n_1 = 497$. For $1000 < n_1 < 10^6$ the minimum of $\exp(L(r,n_1)/n_1)$ is 1.8422.

- For $n_1 \ge 10^6$ the analytic lower estimates of $\log\Pi$ in the author's preprint were recomputed: $0.59197\, n_1$ when $10^6 \le n_1 \le 2\times 10^9$, and $0.59035\, n_1$ when $n_1 > 2\times 10^9$. Both exceed $\log 1.8 \cdot n_1 = 0.58779\, n_1$. Those estimates use the preprint's prime-error bound, in which $1.798158\sqrt{x}$ and $0.00144\, x/\log x$ meet at $x = 641{,}239{,}200.2$. The journal states the coefficient 0.001 and splits this argument at 3×10^9. The constant 1.798158, from Bennett, Martin, O'Bryant and Rechnitzer, was spot-checked for $x \le 2\times 10^8$, and the largest ratio found is 1.79628.

What was not recomputed is that prime bound for $x > 2\times 10^8$, Lemma 4 of Beukers (1980/81), where one choice of δ gives $\Lambda \neq 0$, and Lemma 4.5 of Chudnovsky (1983), which Lemma 6.1 uses.

F Notes on the literature and history

Literature.

- Santos (2020) proves the one- and two-step cases (Propositions 15 and 16) and conjectures that 1 lies on a cycle of C only for $q = 5$ and $q = 2^e - 1$ (Conjecture 3). His Remark 5 describes three or more odd steps as "… much more difficult to analyze and seems intractable"; Theorem 1 settles three steps. The same preprint claims a proof of the Collatz conjecture, which is not accepted; only its elementary propositions are relied on here.

- Crandall (1978) and Franco and Pomerance (1995) study when orbits of $qx+1$ maps avoid 1; Corollary B.3 matches their mechanism.

- Franco's 1990 thesis uses Diophantine approximation for the $qx+1$ problem. Only its title was available, and it is the most important item to check for overlap.

- Steiner (1977, 1981) rules out one-block "circuits" for $q = 3$, 5, 7 with Baker's method, a related use of Diophantine approximation for cycles.

- The deep inputs are Bauer and Bennett (2002, 2018) on restricted irrationality measures, Beukers (1980/81) on $X^2 + D = 2^n$, and Ridout (1958) on approximation with S-unit denominators.

History of the results. The work began with the window (Lemma 8), which pins down q from h and makes a search over h far faster than a search over q. Lower bounds on x and y (Lemmas 4 to 7, preceded by the weaker Proposition B.13, which allowed $y < x$ with $y > (3q)^{1/3}$) then gave finiteness through Ridout's theorem (Proposition B.4), but no bound. Residue-class arguments settled special cases (Propositions B.5 and B.6), and the effective inequality of Bauer and Bennett gave the lower bound on k_1 (Proposition B.7) and most of the case analysis of Lemma 4 (Appendix B). Playing

the window against the near miss for x then gave Theorem 1, once it was noticed that Bauer and Bennett's unrounded exponent lies below the golden ratio. The near-miss principle (Lemma 11) extended this to Theorem 2, Corollaries 12 to 14, and Theorem 3. The digit scan (Appendix E) was computational evidence found along the way. Cycles that need not visit 1 are treated in the companion paper. The final verification found and corrected a dropped constant in an intermediate step of Bauer and Bennett and confirmed their Proposition 5.1 for every $n_1 \geq 497$.

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